PARKINSON'S PROGRESSION MARKERS INITIATIVE

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PPMI CSF Aβ42, t-tau and p-tau181 analysis

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PPMI Project ID: 159

Summary: Over twenty seven hundred CSF samples, collected according to the Parkinson's Progression Markers Initiative (PPMI) Laboratory Manual LP procedure, were analysed for three biomarkers: Amyloid-beta(1-42) (Abeta), total tau (t-tau) and tau phosphorylated at the threonine 181 position (p-tau181), using Elecsys electro-chemi-luminescence immunoassays on the cobas e601 platform (Roche Diagnostics). Among the analysed samples were two pooled quality control CSF samples from PPMI (HC pool and PD pool), two pooled QC samples from UPenn (pools 59 and 60), as well as a number of replicate aliquots from the previous batch of analyses (2017) for the purpose of thoroughly evaluating the lot-to-lot performance of the assay.

Method: CSF samples in this project were analysed using the Elecsys Amyloid-beta(1–42), t-tau and p-tau181 electrochemiluminescence (ECL) immunoassays on a fully automated cobas e601 analyzer (Roche Diagnostics). The Abeta assay has a measurement range of 200 to 1700 pg/mL, the t-tau assay: 80 to 1300 pg/mL, and the p-tau181 assay: 8 to 120 pg/mL.

The analyses were performed between February 22, 2021 and April 16, 2021 in a total of 68 analytical runs over 34 days. The samples were processed at a rate of 40 per run, 80 per day, each day consisting of 2 runs. Single reagent lot was used throughout the duration of the study for all analytes.

A total of 2733 CSF samples were processed in this project, including 2463 PPMI patient samples, 134 PPMI pools, and 136 internal UPenn pool controls. Among the 2463 patient samples, there were 75 “bridging” samples previously analyzed in the 2017 batch. These 75 retests were planned for the purpose of evaluating the assay lot-to-lot performance.

Of the 2463 total PPMI patient samples, 1924 were provided by IU/NCRAD, 490 by BioREP, and 49 by Tel Aviv. IU/NCRAD additionally provided all 134 PPMI QC samples, for a total of 2058 samples from that institution, and a total of 2597 PPMI provided samples.

The “bridging” samples were mixed-in with the regular study samples during the first 10 days of analyses (first 20 runs). Results of the “bridging” experiment are summarized in Figure 1, Figure 2 and in Table 1. Regression equations were obtained using Passing-Bablok method and regression model in the form [2021 result] = Slope * [2017 result] + Intercept.

	Intercept		Slope	
Abeta	8.04	(-169.9 - 153.4)	1.147	(0.9905 - 1.396)
Tau	-3.294	(-14.76 - 8.231)	1.019	(0.9406 - 1.099)
PTau	0.3636	(-0.1786 - 1.011)	0.9347	(0.8846 - 0.9725)

Table 1: Passing-Bablok regression coefficients for the "bridging" experiment.
These values for Tau and PTau met the criteria for acceptable lot-to-lot performance: intercept <50% lower than the lowest reportable value, and slope between 0.90-1.10. In order to achieve complete comparability of the assay results for Abeta we re-scaled the results of all CSF samples to the values measured in 2017, using the regression equation from Table 1. Original Abeta results are reported in the "ABETA_RAW" column in the data report file. Passing-Bablok regression done using R (reference #5).

The PPMI pools: "HC Pool" and "PD Pool", were Healthy-Control-like and Parkinson's-Disease-like pools, respectively. The internal pool "59" was prepared from leftover CSF samples of healthy controls and Mild Cognitive Impairment patients, while pool "60" was prepared from leftover CSF samples of Alzheimer's Disease patients.

Pools and internal QCs were processed at the end of each run, 2 aliquots per day – throughout the entire study. Results of these experiments are summarized in Table 2, and in Figure 3.

	group	N	Mean	SD	CV	Median	LCI	UCI
Abeta	BRL Pool 59	68	895	49.29	5.51	903.8	778.8	961.9
	BRL Pool 60	68	524.3	20.03	3.82	524.8	485.6	562.2
	PPMI HC Pool	67	842.4	41.27	4.9	846.5	749.1	909.4
	PPMI PD Pool	67	599.2	34.62	5.78	596	537.2	656.7
t-tau	BRL Pool 59	68	182.4	4.773	2.62	181.4	175.6	191.2
	BRL Pool 60	68	314.8	9.114	2.9	312.9	300.2	330.6
	PPMI HC Pool	67	135.1	4.439	3.29	134.6	127.3	144.1
	PPMI PD Pool	67	116.1	3.423	2.95	116.1	109.3	122
p-tau181	BRL Pool 59	68	15.2	0.6291	4.14	15.11	14.37	16.61
	BRL Pool 60	68	24.34	0.6726	2.76	24.29	23.08	26.06
	PPMI HC Pool	67	13.74	0.6413	4.67	13.66	12.83	15.35
	PPMI PD Pool	67	11.98	0.6243	5.21	11.91	11.17	13.89

Table 2: Overall pool summary

In the data report file, in 288 instances the values of Aβ42 was above the 1700 pg/mL limit, and in one instance below the 200 pg/mL limit. In 78 instances the t-tau value was below 80 pg/mL, and in 170 instances the value of p-tau181 was below the 8 pg/mL limit. All these values are reported numerically with the value of the limit, and annotated in the "Note" column.

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In addition, 15 results (0.58%) were lost due to analyzer conveyor belt malfunction, and 7 more samples (0.27%) were not analyzable due to aspiration flag. These records are marked in the data report file as "technical problems" - in the "Note" column.

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5. R "mcr" package, version 1.2.2.

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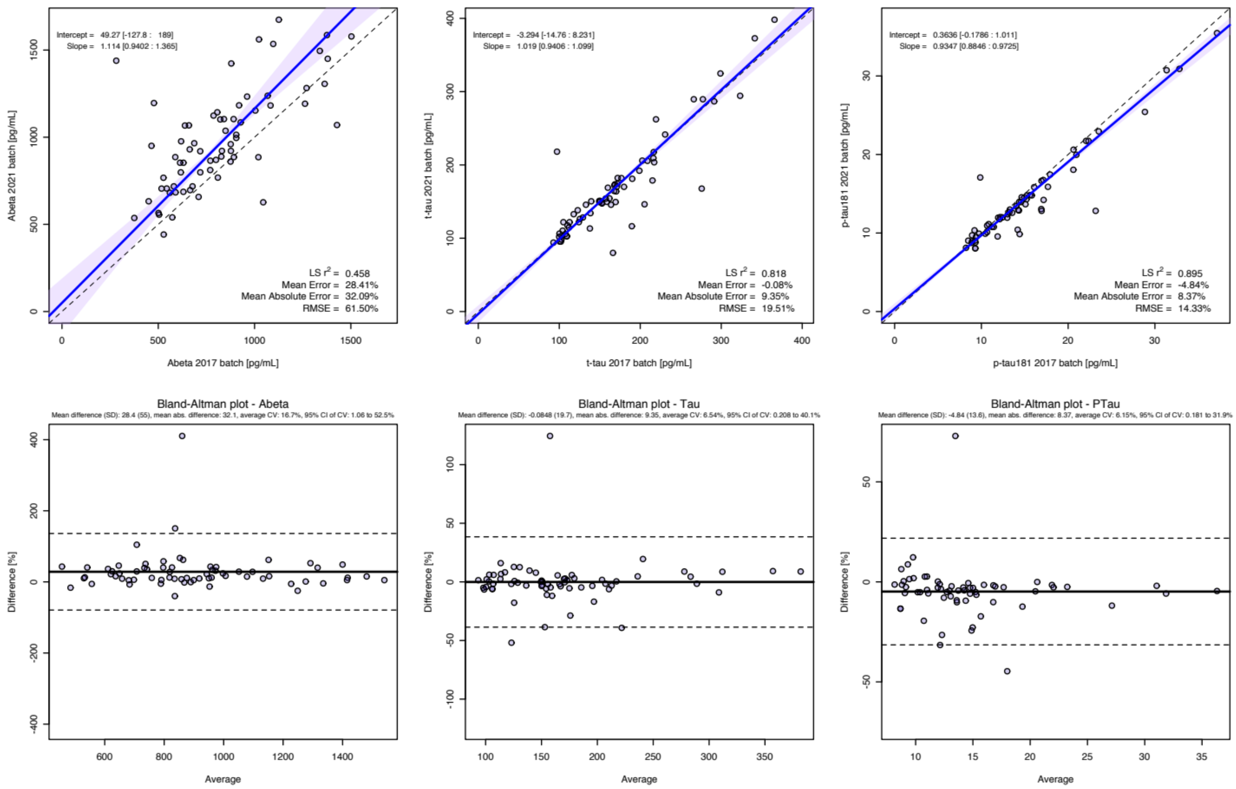

Figure 1: Summary of test-retest results for Abeta, Tau and PTau assays

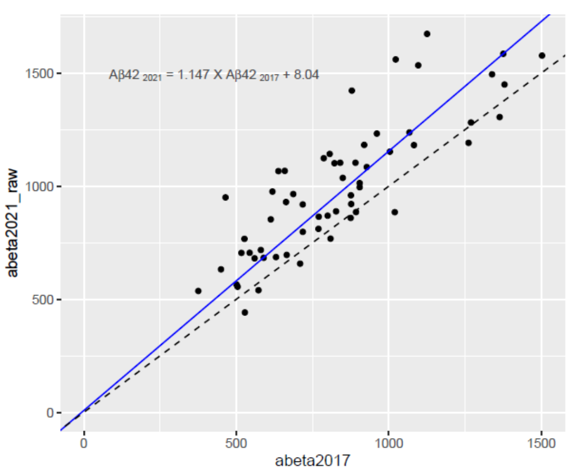

Figure 2: Test re-test results for Aβ4 2 with 4 outliers removed: Pt 3066, V02; pt 3277, V04; pt. 3565, V06 and pt. 3613, V08. In addition 10 abeta42 results were measured at 1,700 pg/mL so at the upper limit of quantification and were therefore not included in the test/re-test exercise. there were no values for 2 samples for the 2017 batch abeta42 results and therefore those were excluded as well.

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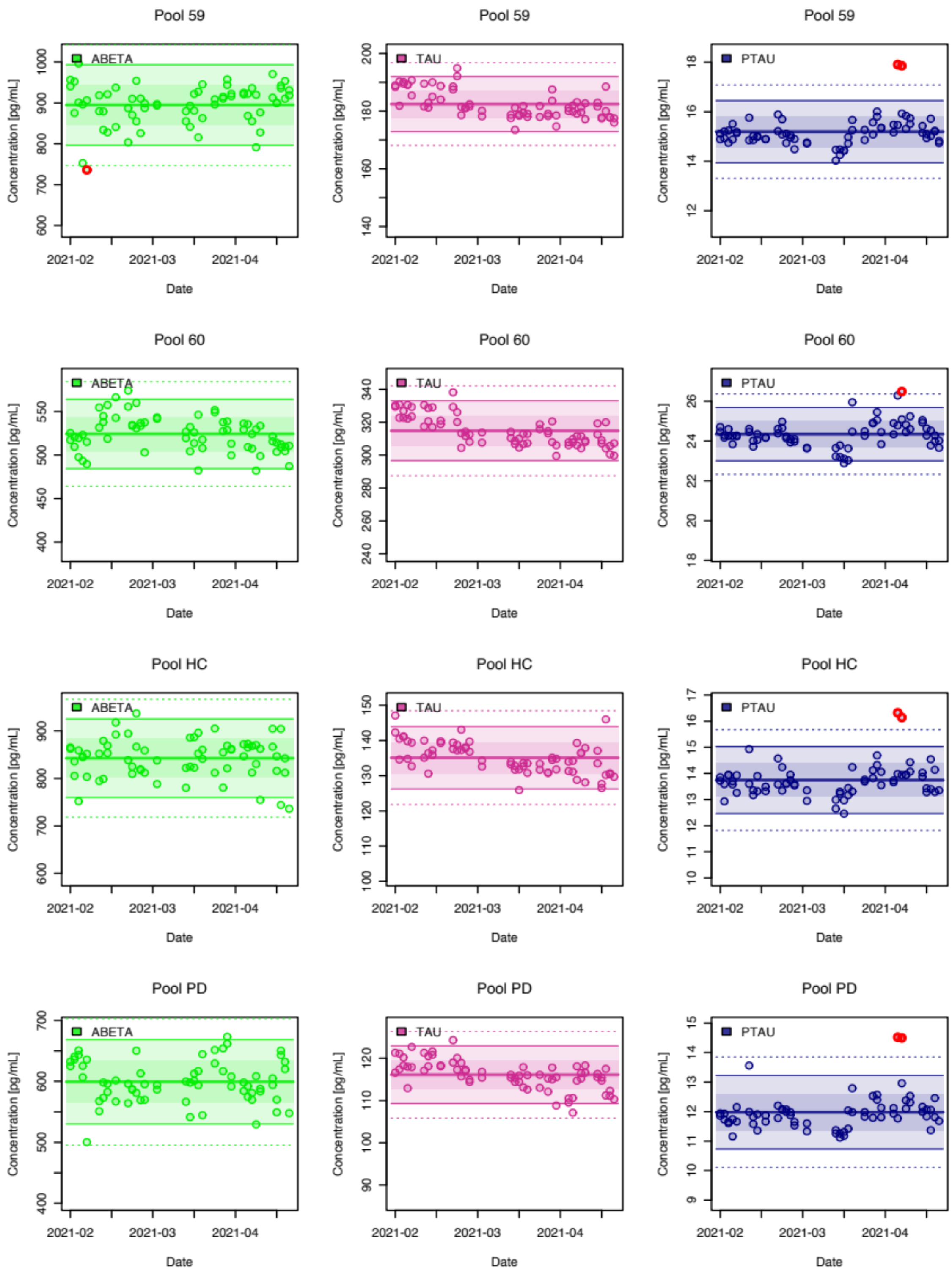

Figure 3: QC pools data for 68 analytical runs performed during 34 days.